
MATH2015 – Perturbation Method

Matched Asymptotic Method

This document studies the boundary value problem

$$\varepsilon y''(x) - 2(x^2 + 1)y'(x) + 1 = 0, \quad x \in [0, 1], \quad 0 < \varepsilon \ll 1, \quad (1)$$

subject to the following boundary conditions

$$y(0) = 0, \quad y(1) = 0. \quad (2)$$

Since the highest-order derivative in Equation 1 is multiplied by the small parameter ε , the problem is singularly perturbed. It is therefore natural to seek an approximate solution by means of matched asymptotic expansions.

The aim of this document is to derive an analytical approximation consisting of the leading-order term and its first correction, and then to construct the corresponding composite expansion. To this end, the location and thickness of the boundary layer are first identified in section 1. The outer and inner expansions are then derived in section 2 and section 3, respectively. These expansions are matched in section 4 in order to obtain a uniformly valid composite approximation.

Finally, a numerical solution is computed and compared with the asymptotic approximation in section 5 in order to assess the accuracy of the analytical construction.

1 Location and thickness of the boundary layer

We begin by determining the location and thickness of the boundary layer. Setting $\varepsilon = 0$ in Equation 1 reduces the equation to a first-order problem, so the outer solution can satisfy only one boundary condition. The remaining condition must therefore be imposed through a boundary layer.

For a singularly perturbed equation of the form

$$\varepsilon y'' + a(x)y' + \dots = 0,$$

the sign of $a(x)$ indicates where the layer is located. In the present case,

$$a(x) = -2(x^2 + 1) < 0, \quad x \in [0, 1], \quad (3)$$

so the boundary layer is located near $x = 1$.

To determine its thickness, we first introduce a stretched variable of the form

$$X = \frac{x}{\delta},$$

where δ denotes the width of the boundary layer. Since the layer is attached to the right endpoint $x = 1$, it is more convenient to use the shifted stretched coordinate

$$X = \frac{1 - x}{\delta}.$$

Writing the inner solution as

$$y(x) = Y(X),$$

the derivatives become

$$\frac{dy}{dx} = -\frac{1}{\delta}Y'(X), \quad \frac{d^2y}{dx^2} = \frac{1}{\delta^2}Y''(X).$$

Substituting into Equation 1 gives

$$\frac{\varepsilon}{\delta^2}Y'' + \frac{2(x^2 + 1)}{\delta}Y' + 1 = 0.$$

A nontrivial dominant balance in the boundary layer is obtained by requiring

$$\frac{\varepsilon}{\delta^2} \sim \frac{1}{\delta},$$

hence

$$\delta = O(\varepsilon).$$

Therefore, the appropriate inner variable is

$$X = \frac{1 - x}{\varepsilon},$$

and the boundary layer has thickness of order ε .

2 Outer expansion

We now derive the outer expansion, valid away from the boundary layer near $x = 1$. We seek a regular asymptotic expansion of the form

$$y(x) \sim y_0(x) + \varepsilon y_1(x) + \cdots. \quad (4)$$

Substituting Equation 4 into Equation 1 yields

$$\varepsilon(y_0'' + \varepsilon y_1'' + \cdots) - 2(x^2 + 1)(y_0' + \varepsilon y_1' + \cdots) + 1 = 0. \quad (5)$$

Collecting terms of like powers of ε gives a hierarchy of equations for $y_0, y_1, \dots$

Leading order

At order $O(1)$, Equation 5 gives

$$-2(x^2 + 1)y_0'(x) + 1 = 0,$$

so that

$$y_0'(x) = \frac{1}{2(x^2 + 1)}.$$

Integrating, we obtain

$$y_0(x) = \frac{1}{2} \arctan x + C_0.$$

Since the boundary layer is located at $x = 1$, the outer solution must satisfy the left boundary condition. Imposing $y_0(0) = 0$ gives $C_0 = 0$, and therefore

$$y_0(x) = \frac{1}{2} \arctan x. \quad (6)$$

At $x = 1$,

$$y_0(1) = \frac{1}{2} \arctan(1) = \frac{\pi}{8} \neq 0,$$

which confirms that the outer expansion cannot satisfy the right boundary condition and must be considered with an inner correction near $x = 1$.

First correction

At order $O(\varepsilon)$, Equation 5 gives

$$y_0''(x) - 2(x^2 + 1)y_1'(x) = 0,$$

hence

$$y_1'(x) = \frac{y_0''(x)}{2(x^2 + 1)}. \quad (7)$$

From Equation 6,

$$y_0'(x) = \frac{1}{2(1 + x^2)}, \quad y_0''(x) = -\frac{x}{(1 + x^2)^2}.$$

Substituting into Equation 7 gives

$$y_1'(x) = -\frac{x}{2(1 + x^2)^3}.$$

Integrating, we find

$$y_1(x) = \frac{1}{8(1 + x^2)^2} + C_1.$$

The constant C_1 is determined by imposing the left boundary condition order by order, namely $y_1(0) = 0$. This gives

$$0 = \frac{1}{8} + C_1, \quad \text{so} \quad C_1 = -\frac{1}{8}.$$

Therefore,

$$y_1(x) = \frac{1}{8(1 + x^2)^2} - \frac{1}{8}. \quad (8)$$

Hence, the outer expansion up to and including the first correction is

$$y_{\text{out}}(x) = \frac{1}{2} \arctan x + \varepsilon \left(\frac{1}{8(1 + x^2)^2} - \frac{1}{8} \right). \quad (9)$$

3 Inner expansion

We now derive the inner expansion in the boundary layer near $x = 1$. As established in section 1, the appropriate stretched variable is

$$X = \frac{1 - x}{\varepsilon}, \quad (10)$$

and we introduce the inner unknown $Y(X)$ through

$$y(x) = Y(X).$$

By the chain rule,

$$\frac{dy}{dx} = -\frac{1}{\varepsilon} Y'(X), \quad \frac{d^2y}{dx^2} = \frac{1}{\varepsilon^2} Y''(X).$$

Substituting these expressions into Equation 1, and using $x = 1 - \varepsilon X$, gives

$$\varepsilon \frac{1}{\varepsilon^2} Y'' - 2((1 - \varepsilon X)^2 + 1) \left(-\frac{1}{\varepsilon} Y' \right) + 1 = 0,$$

that is,

$$\frac{1}{\varepsilon} Y'' + \frac{2}{\varepsilon} ((1 - \varepsilon X)^2 + 1) Y' + 1 = 0.$$

Multiplying by ε , we obtain the inner equation in the form

$$Y'' + 2((1 - \varepsilon X)^2 + 1)Y' + \varepsilon = 0. \quad (11)$$

This is now a regular perturbation problem in the inner variable, so we seek an expansion of the form

$$Y(X) \sim Y_0(X) + \varepsilon Y_1(X) + \dots. \quad (12)$$

Moreover,

$$2((1 - \varepsilon X)^2 + 1) = 2(1 - 2\varepsilon X + \varepsilon^2 X^2 + 1) = 4 - 4\varepsilon X + O(\varepsilon^2).$$

Substituting this into Equation 11 yields

$$Y'' + (4 - 4\varepsilon X + O(\varepsilon^2))Y' + \varepsilon = 0.$$

Leading order

Substituting Equation 12 into the inner equation and collecting the terms of order $O(1)$ gives

$$Y_0'' + 4Y_0' = 0.$$

The general solution is

$$Y_0(X) = A_0 e^{-4X} + B_0.$$

At the right boundary, $x = 1$ corresponds to $X = 0$, and the boundary condition $y(1) = 0$ therefore becomes

$$Y_0(0) = 0.$$

Hence,

$$A_0 + B_0 = 0,$$

so that

$$Y_0(X) = B_0(1 - e^{-4X}).$$

The constant B_0 is determined by matching with the outer solution. In the matching region, $x \rightarrow 1^-$ and $X \rightarrow +\infty$, so we require

$$\lim_{X \rightarrow +\infty} Y_0(X) = \lim_{x \rightarrow 1^-} y_0(x).$$

Using Equation 6, we obtain

$$\lim_{x \rightarrow 1^-} y_0(x) = \frac{1}{2} \arctan(1) = \frac{\pi}{8},$$

while

$$\lim_{X \rightarrow +\infty} Y_0(X) = B_0.$$

Therefore,

$$B_0 = \frac{\pi}{8},$$

and thus

$$Y_0(X) = \frac{\pi}{8}(1 - e^{-4X}). \quad (13)$$

First correction

At order $O(\varepsilon)$, we obtain

$$Y_1'' + 4Y_1' - 4XY_0' + 1 = 0.$$

From Equation 13,

$$Y_0'(X) = \frac{\pi}{2}e^{-4X},$$

so the equation becomes

$$Y_1'' + 4Y_1' - 2\pi X e^{-4X} + 1 = 0.$$

Introducing

$$Z(X) = Y_1'(X),$$

this reduces to the first-order equation

$$Z' + 4Z = 2\pi X e^{-4X} - 1.$$

Multiplying by the integrating factor e^{4X} gives

$$\frac{d}{dX}(e^{4X}Z) = 2\pi X - e^{4X}.$$

Integrating, we find

$$e^{4X}Z = \pi X^2 - \frac{1}{4}e^{4X} + C,$$

hence

$$Y_1'(X) = \pi X^2 e^{-4X} - \frac{1}{4} + C e^{-4X}.$$

Integrating once more,

$$Y_1(X) = -\frac{X}{4} + D - e^{-4X} \left(\frac{\pi}{4}X^2 + \frac{\pi}{8}X + \frac{\pi}{32} + \frac{C}{4} \right).$$

It remains to determine the constants C and D . For this, we match the inner expansion with the outer expansion written in terms of the inner variable X . As $x = 1 - \varepsilon X$,

$$y_0(1 - \varepsilon X) = y_0(1) - \varepsilon X y_0'(1) + O(\varepsilon^2), \quad y_1(1 - \varepsilon X) = y_1(1) + O(\varepsilon).$$

Therefore,

$$y_{\text{out}}(1 - \varepsilon X) \sim \frac{\pi}{8} + \varepsilon(-X y_0'(1) + y_1(1)).$$

Using Equation 6 and Equation 8,

$$y_0'(1) = \frac{1}{2(1+1^2)} = \frac{1}{4}, \quad y_1(1) = \frac{1}{8(1+1^2)^2} - \frac{1}{8} = -\frac{3}{32}.$$

Hence the matching condition at order $O(\varepsilon)$ is

$$Y_1(X) \sim -\frac{X}{4} - \frac{3}{32}, \quad X \rightarrow +\infty.$$

Since the exponential term vanishes as $X \rightarrow +\infty$, this immediately gives

$$D = -\frac{3}{32}.$$

The boundary condition at $x = 1$ must also be imposed order by order in the inner expansion, so

$$Y_1(0) = 0.$$

Evaluating the general expression at $X = 0$ gives

$$0 = -\frac{3}{32} - \left(\frac{\pi}{32} + \frac{C}{4} \right),$$

hence

$$\frac{C}{4} = -\frac{\pi + 3}{32}.$$

Substituting this into the expression for Y_1 yields

$$Y_1(X) = -\frac{X}{4} - \frac{3}{32} + e^{-4X} \left(\frac{3}{32} - \frac{\pi}{8}X - \frac{\pi}{4}X^2 \right). \quad (14)$$

Therefore, the inner expansion up to first order is

$$Y(X) \sim \frac{\pi}{8}(1 - e^{-4X}) + \varepsilon \left[-\frac{X}{4} - \frac{3}{32} + e^{-4X} \left(\frac{3}{32} - \frac{\pi}{8}X - \frac{\pi}{4}X^2 \right) \right]. \quad (15)$$

Equivalently, in terms of x ,

$$y_{\text{in}}(x) \sim \frac{\pi}{8} (1 - e^{-4(1-x)/\varepsilon}) + \varepsilon \left[-\frac{1-x}{4\varepsilon} - \frac{3}{32} + e^{-4(1-x)/\varepsilon} \left(\frac{3}{32} - \frac{\pi(1-x)}{8\varepsilon} - \frac{\pi(1-x)^2}{4\varepsilon^2} \right) \right]. \quad (16)$$

4 Matching and composite expansion

To construct a uniformly valid approximation, we first identify the common part of the outer and inner expansions. This overlap is obtained by taking $X \rightarrow +\infty$ in the inner expansion, or equivalently $x \rightarrow 1^-$ in the outer expansion.

From Equation 13 and Equation 14,

$$Y_0(X) \sim \frac{\pi}{8}, \quad Y_1(X) \sim -\frac{X}{4} - \frac{3}{32}, \quad X \rightarrow +\infty.$$

Therefore, the overlap expansion is

$$y_{\text{matching}} = \frac{\pi}{8} + \varepsilon \left(-\frac{X}{4} - \frac{3}{32} \right), \quad X = \frac{1-x}{\varepsilon}.$$

The composite expansion is then defined by

$$y_{\text{comp}} = y_{\text{out}} + y_{\text{in}} - y_{\text{matching}}.$$

Leading-order composite solution

At leading order, this gives

$$y_{\text{comp}}^{(0)}(x) = \frac{1}{2} \arctan x + \frac{\pi}{8}(1 - e^{-4X}) - \frac{\pi}{8},$$

that is,

$$y_{\text{comp}}^{(0)}(x) = \frac{1}{2} \arctan x - \frac{\pi}{8} e^{-4(1-x)/\varepsilon}. \quad (17)$$

Composite solution including the first correction

At order $O(\varepsilon)$, subtracting the overlap removes the non-decaying part of the inner correction, namely

$$-\frac{X}{4} - \frac{3}{32},$$

so only the exponentially small remainder is retained:

$$e^{-4X} \left(\frac{3}{32} - \frac{\pi}{8}X - \frac{\pi}{4}X^2 \right).$$

Hence the uniformly valid composite approximation up to first order is

$$y_{\text{comp}}(x) = \frac{1}{2} \arctan x - \frac{\pi}{8} e^{-4(1-x)/\varepsilon} + \varepsilon \left[\frac{1}{8(1+x^2)^2} - \frac{1}{8} + e^{-4(1-x)/\varepsilon} \left(\frac{3}{32} - \frac{\pi}{8} \frac{1-x}{\varepsilon} - \frac{\pi}{4} \frac{(1-x)^2}{\varepsilon^2} \right) \right].$$

This is the matched asymptotic composite expansion, including the leading-order term and its first correction.

5 Comparison with numerical results

In this section, the accuracy of the composite asymptotic approximation is assessed by comparison with a numerical reference solution. The latter is computed using the Python script provided with the homework, based on the function `solve_bvp` from the `scipy.integrate` library.

To evaluate the quality of the asymptotic approximation, it is natural to compare the solutions for several values of the small parameter ε . In what follows, we consider $\varepsilon \in \{0.1, 0.05, 0.01\}$. For larger values, such as $\varepsilon = 1$, the assumption $0 < \varepsilon \ll 1$ is no longer valid, so the perturbation approach is not expected to provide an accurate approximation. Conversely, for very small values such as $\varepsilon = 10^{-5}$, a uniform mesh with only a few tens of grid points is generally insufficient to resolve the sharp variation inside the boundary layer. These two limitations are illustrated in Figure 1.

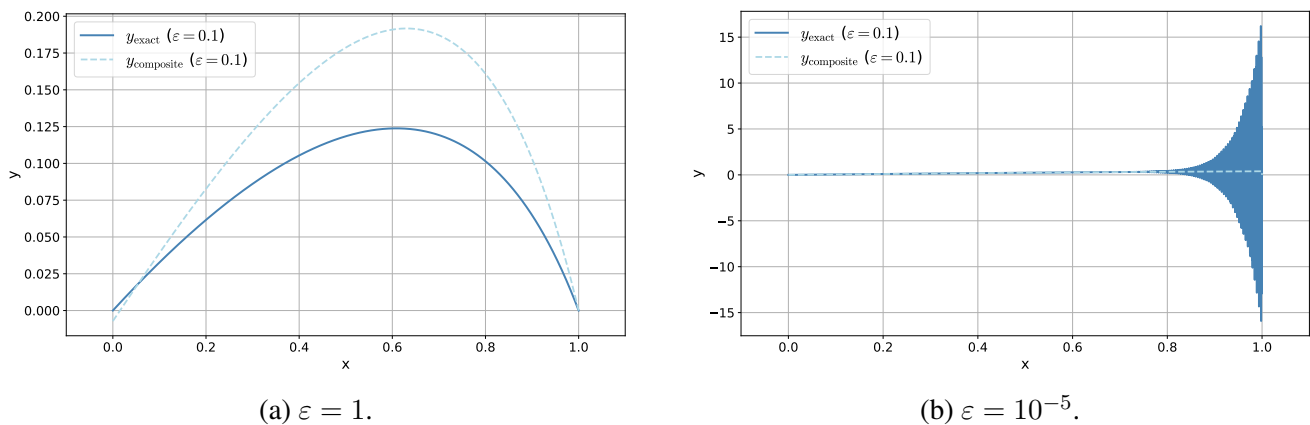


Figure 1: Illustration of two limiting cases: failure of the asymptotic approximation for large ε , and insufficient spatial resolution of the numerical solution for very small ε .

In Figure 2, the leading-order composite approximation is compared with the numerical solution for several values of ε . The same comparison is performed in Figure 3 for the composite approximation including the $O(\varepsilon)$ correction. The results show that the leading-order approximation already captures the boundary-layer structure, but still exhibits visible discrepancies. By contrast, the first-order composite expansion provides a significantly more accurate approximation over the whole interval.

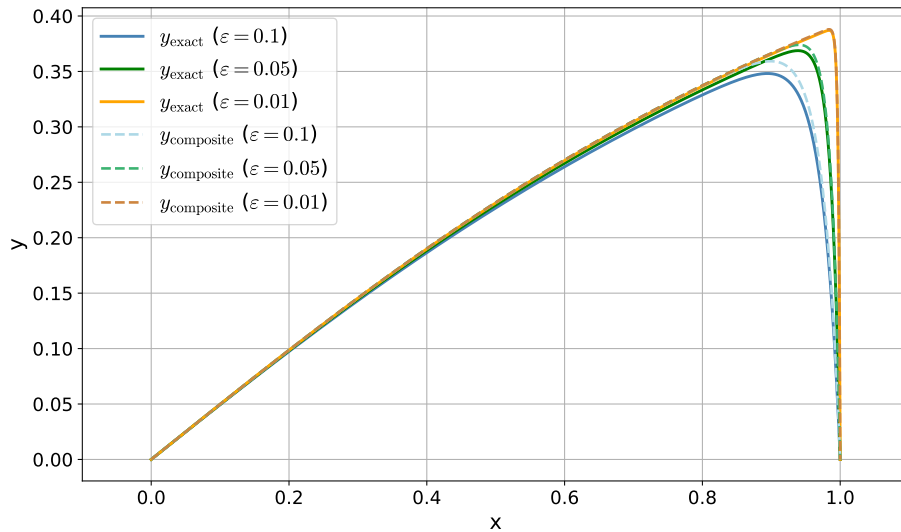


Figure 2: Comparison between the numerical solution and the leading-order composite approximation for several values of ε .

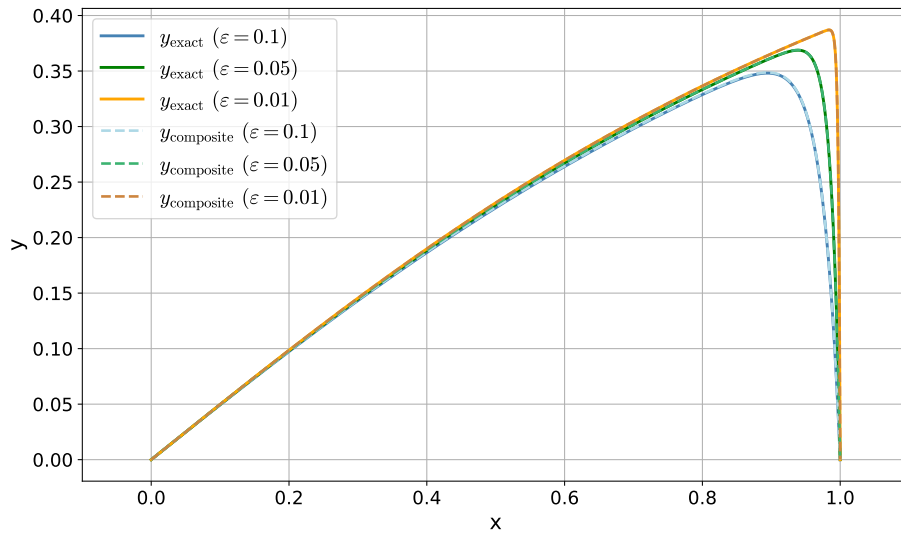


Figure 3: Comparison between the numerical solution and the composite approximation including the $O(\varepsilon)$ correction for several values of ε .

5.1 Quantitative error analysis

In order to assess more precisely the quality of the two composite approximations, a quantitative error analysis is now performed. For an approximation u_{app} of the exact solution u_{exact} , the error

$$e(x) = u_{\text{app}}(x) - u_{\text{exact}}(x)$$

is evaluated on a fine grid, and the following norms are considered:

$$\|e\|_{L^\infty} = \max_{x \in [0,1]} |e(x)|, \quad \|e\|_{L^2} = \left(\int_0^1 |e(x)|^2 dx \right)^{1/2}.$$

Since the amplitude of the exact solution varies slightly with ε , the corresponding relative errors are also reported:

$$\frac{\|e\|_{L^\infty}}{\|u_{\text{exact}}\|_{L^\infty}}, \quad \frac{\|e\|_{L^2}}{\|u_{\text{exact}}\|_{L^2}}.$$

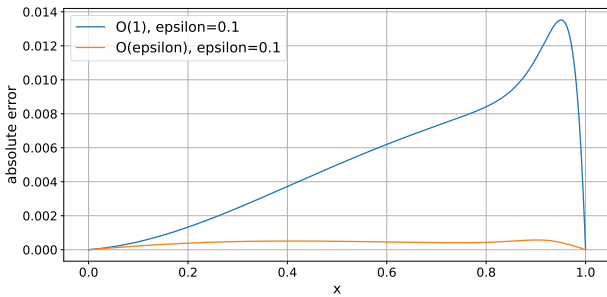
The numerical values obtained are given in Table 1. They clearly show that the corrected composite solution is significantly more accurate than the leading-order one for all tested values of ε . In both the L^∞ and L^2 norms, the error decreases by roughly one to two orders of magnitude when passing from the $O(1)$ composite approximation to the $O(\varepsilon)$ corrected approximation.

ε	Model	$\ e\ _{L^\infty}$	$\ e\ _{L^2}$	Relative L^∞	Relative L^2
0.1	$O(1)$	1.3523×10^{-2}	6.3105×10^{-3}	3.8844×10^{-2}	2.7458×10^{-2}
0.1	$O(\varepsilon)$	5.7191×10^{-4}	4.2800×10^{-4}	1.6427×10^{-3}	1.8623×10^{-3}
0.05	$O(1)$	6.7521×10^{-3}	2.9508×10^{-3}	1.8310×10^{-2}	1.2351×10^{-2}
0.05	$O(\varepsilon)$	1.5345×10^{-4}	1.1566×10^{-4}	4.1612×10^{-4}	4.8413×10^{-4}
0.01	$O(1)$	1.3459×10^{-3}	5.5351×10^{-4}	3.4773×10^{-3}	2.2508×10^{-3}
0.01	$O(\varepsilon)$	9.6340×10^{-6}	4.9453×10^{-6}	2.4891×10^{-5}	2.0110×10^{-5}

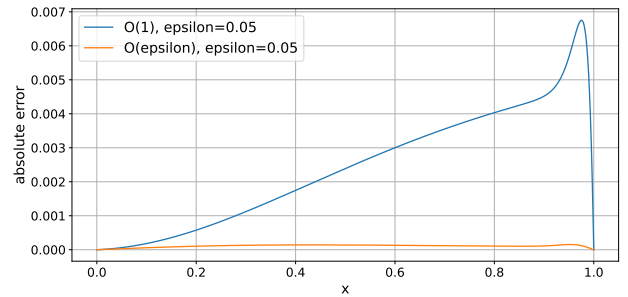
Table 1: Absolute and relative error norms for the leading-order and corrected composite solutions.

These results are consistent with the visual comparisons of Figures 2 and 3. The leading-order composite solution already captures the global structure of the exact solution, but a non-negligible discrepancy remains, especially close to the right boundary layer. By contrast, the corrected composite solution follows the exact solution much more closely over the whole interval.

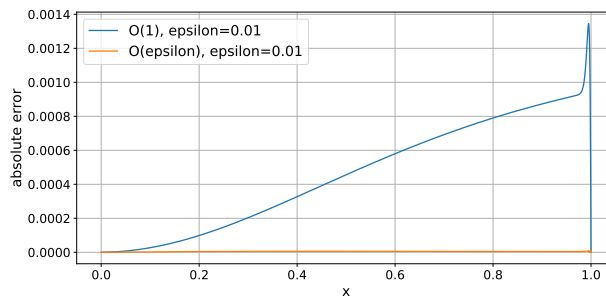
This behaviour is confirmed by the pointwise absolute error profiles shown in Figure ???. In each case, the error associated with the leading-order approximation increases as one approaches the boundary layer, whereas the corrected approximation remains much smaller throughout the domain. One also observes that the largest errors are attained near $x = 1$, which is precisely where the rapid variation of the solution takes place.



(a) $\varepsilon = 0.1$.



(b) $\varepsilon = 0.05$.



(c) $\varepsilon = 0.01$.

Finally, the location of the maximum error also provides useful information. For the leading-order approximation, the maximum error is located very close to the right endpoint, and this concentration becomes even stronger as ε decreases. The same phenomenon is observed for the corrected approximation, although with much smaller amplitude. This confirms that the main source of discrepancy lies in the boundary layer region, and that the $O(\varepsilon)$ correction substantially improves the approximation precisely where it is most needed.

For completeness, the locations of the maximum error are reported below:

$$\begin{aligned}\varepsilon = 0.1, \quad O(1) : x &= 0.950390, \\ \varepsilon = 0.1, \quad O(\varepsilon) : x &= 0.900780, \\ \varepsilon = 0.05, \quad O(1) : x &= 0.975195, \\ \varepsilon = 0.05, \quad O(\varepsilon) : x &= 0.954391, \\ \varepsilon = 0.01, \quad O(1) : x &= 0.994999, \\ \varepsilon = 0.01, \quad O(\varepsilon) : x &= 0.999600.\end{aligned}$$

Overall, both the graphical and quantitative analyses lead to the same conclusion: the corrected composite solution of order ε provides a much better approximation to the exact numerical solution than the leading-order composite approximation, especially in the boundary layer region.

6 Use of artificial intelligence tools

Concerning the first part of this homework, namely the analytical solution, I used some AI tools to help clarify a few concepts from the theoretical lectures. During the analytical development, I made some conceptual mistakes and initially followed unnecessarily complicated lines of reasoning, which I was later able to correct with the help of AI. However, these tools were never used to generate a complete solution to the problem, they were only used as support when I was stuck or needed clarification.

For the second part, concerning the numerical method, AI was only used to provide a short script for the computation of the error, as explained at the beginning of the notebook attached to this document.